

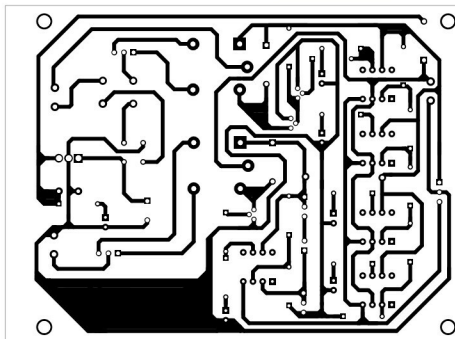


Fig. 3: Actual-size PCB for automatic light and overcharging control

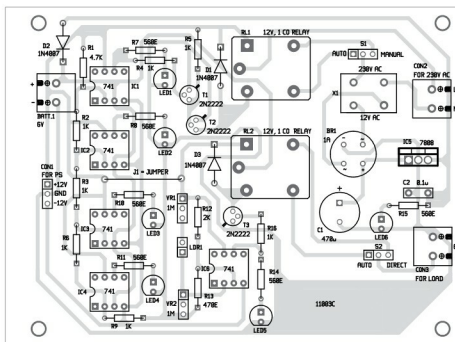


Fig. 4: Components layout for the PCB

In direct mode, the load can be switched on/off without LDR1. In auto mode, the load is controlled by LDR1. During night, the load will be 'on.' In daytime, the load will be 'off.' For detailed functions of switches S1 and S2, refer Table II.

Construction and testing

An actual-size PCB for automatic light and overcharging control with voltage-level indicator is shown in Fig. 3 and its components layout in Fig. 4. Assemble the circuit and enclose it in a suitable box. Connect $\pm 12V$ power supply across CON1, 230V AC across CON2 and the load across CON3. Affix LDR1 at a suitable location such that sunlight falls on it directly.

For testing purpose, use a variable voltage source with a potentiometer in place of the rechargeable battery (BATT.1). Connect one terminal of the potentiometer to +6V and the other terminal to ground. Connect the wiper or mid terminal of the potentiometer to pin 3 of IC1 through IC4. To indicate battery discharging, increase potentiometer resistance. Check the status of LED1 through LED4. Also check switches S1 and S2 in both the modes. **EFY**

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